

LC 994: Rotting Oranges

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Asked in: Flipkart, Myntra, Zepto, Deutsche

Problem Understanding

2

Rotten Orange

1

Fresh Orange

0

Empty

Logic & Approaches

Wrong Approach

- S1: Rotten 1 → BFS
- S2: Rotten 2 → BFS
- S3: Rotten 3 → BFS

Correct Approach

At $t = 0$, all rotten oranges exist, so they spread **together**.

Standard Template

S1: push all sources

S2:

```
while(!q.empty()) {  
    int currsz = q.size();  
    while(currsz--) {  
        // process the current minute  
    }  
    minute++;  
}
```

also keep a track of no. of fresh oranges.
If > 0 , return -1.

Can DFS work?

Technically yes, but not naturally. It can't simulate, everyone spreads naturally.

C++ Implementation

```
class Solution {
public:
    int orangesRotting(vector<vector<int>>& grid) {
        int n = grid.size();
        int m = grid[0].size();

        vector<pair<int,int>> dir = {{1,0}, {-1,0}, {0,1}, {0,-1}};
        queue<pair<int,int>> q;

        int fresh = 0;
        for(int i = 0; i < n; i++){
            for(int j = 0; j < m; j++){
                if(grid[i][j] == 2){
                    q.push({i,j});
                }
                else if(grid[i][j] == 1){
                    fresh++; // total fresh count which by end of bfs must
turn 0 otherwise -1
                }
            }
        }

        if(fresh == 0) return 0;

        int ans = 0;
        while(!q.empty()){
            int currsz = q.size(); // poore batch ko process krna he for
that minute
            while(currsz--){

                int r = q.front().first;
                int c = q.front().second;
                q.pop();

                for(auto it : dir){
                    int nr = r + it.first;
                    int nc = c + it.second;
                    if(nr >= 0 && nc >= 0 && nr < n && nc < m){
                        if(grid[nr][nc] == 1){
                            grid[nr][nc] = 2;
                            q.push({nr,nc});
                            fresh--;
                        }
                    }
                }
            }
        }

        ans++; // ek batch process hote hi
    }
};
```

```
}  
  
if(fresh > 0) return -1;  
else return ans - 1;    // -1 ka intuition dry run krke hi aayega  
}  
};
```